MILITARY SPECIFICATION

# ROLLER, MOTORIZED: COMPACTOR, VIBRATORY, ARTICULATED, DIESEL ENGINE DRIVEN

This specification is approved for use by the Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

2.1 This specification covers one type vibratory compactor roller. This roller shall be diesel engine driven and pneumatic tired and shall have an articulated frame with two interchangeable drums.

## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-F-800 - Fuel, Oil, Diesel

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment)

MIL-P-514 - Plate, Identification, Instruction and Marking Blank

MIL-T-704 - Treatment and Painting of Materiel

MIL-A-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service

MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing it the end of this document or by letter.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

| | |
|---|---|
| MIL-R-3075 | Roller, Motorized, Road, Diesel or Gasoline-Engine-Driven, Packaging of |
| MIL-T-5624 | Turbine Fuel, Aviation, Grade JP-4 and JP-5 |
| MIL-G-10924 | Grease, Automotive and Artillery |
| MIL-M-43719 | Marking Materials, and Markers, Adhesive Elastomeric, Pigmented, General Specification For |
| MIL-L-46167 | Lubricating Oil, Internal Combustion Engine, Arctic |
| MIL-E-52798 | Enamel, Alkyd, Camouflage |
| MIL-C-62122 | Cable Assembly, Slave, Electrical, Connectors, Plug and Receptacle |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-130 | - | Identification Marking of U. S. Military Property |
| MIL-STD-209 | - | Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment |
| MIL-STD-461 | - | Electromagnetic Interference Characteristic Requirements for Equipment |
| MIL-STD-462 | - | Electromagnetic Interference Characteristics, Measurement of |
| MS35000-3 | - | Battery, Storage, Lead-Acid, Waterproof |
| MS39060 | - | Switch, Rotary, 28 Volt DC, Waterproof |
| MS51118 | - | Pintle Assembly, Towing - 40,000 lbs capacity, Manual Release |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, N.W., Washington, D.C. 20036. )

THE TIRE AND RIM ASSOCIATION, INC.

Tire & Rim Year Book

(Application for copies should be addressed to the Tire & Rim Association Incorporated, 3200 W. Market Street, Akron, Ohio 44313. )

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-1557-70 - Mositure-Density Relations of Soils using (4.5kg) Rammer and 18-in (45.7mm) Drop, Methods of Test For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ENVIRONMENTAL PROTECTION (EPA)

40 CFR, Part 85 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent for Documents, U.S. Government Printing Office, Washtington, D. C. 20402.)

DEPARTMENT OF LABOR (DOL)

29 CRF 1926 - Occupational Safety and Health Administration (OSHA) Standards for Construction Jobs.



Source: https://assist.dla.mil -- Downloaded: 2026l0-20T22:43Z
Check the source to verify that this is the current version before use.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Document, U. S. Government Printing Office, Washinton, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

Test Specifications - Standard T99

(Application for copies should be addressed to the American Association of State Highway and Transportation Officials, 444 North Capitol Street, Washington, D.C. 20001.)

FACTOR MUTUAL RESEARCH

OE5A0.AF (7813) Fuel System

(Application for copies should be addressed to Factory Mutual Research, 1151 Boston-Providence Turnpike, Norwood, Massachussetts 02062.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J106 Soil Type and Strength Classification

J429 Mechancial and Material Requirements for External Threaded Fasteners

J525 Welded and Cold Drawn Low Carbon Steel Annealed for Bending and Flaring

J816b Engine Test Code

J919b Operator Sound Level Measurement Procedure for Powered Mobile Construction Machinery

J1136 Minimum Performance Criteria for Braking Systems for Self-Propelled, Smooth Steel Wheel, Pneumatic Tired or Multiple Projection Steel Wheel Rollers and Compactors

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

* 3.1 First article. One complete roller including one smooth vibratory drum, one padfoot vibratory drum and any drum changeover equipment required conforming to all requirements of this specification, shall be submitted for first article inspection and tests. Approval of the first article shall not relieve the supplier of the responsibility of furnishing equipment in accordance with this specification. The first article shall remain at the manufacturer's plant as a sample and shall be delivered as the last unit of the initial contract.

3.2 Material. Except as specifically specified herein, all materials, components, and design shall be identical to that currently sold by the manufacturer. Rubber products furnished, including tires, shall not exceed eight calendar quarters in age from cure date to date of acceptance of each roller by the Government.

3.2.1 Fungus and mositure resistance. All electrical circuitry connections and components of all electrical systems, except those noted below shall be coated with varnish conforming to MIL-V-173:

(a) Components and elements inherently inert to fungi or hermetically sealed.

(b) Components and elements whose operation would be adversely affected by the application of varnish.

* 3.3 Design. The roller shall be a diesel engine driven, pneumatic tired power unit coupled through an articulated, hydraulically actuated hinge point steering system to either of two interchangeable drum/drum and frame assemblies, one that shall contain a single smooth vibratory drum and one that shall contain a single padfoot vibratory drum. The system shall be designed to provide for simple and quick interchangeability of drum/frame modules to match soil material and site conditions. Each drum/frame module shall be equipped with a drum drive system to assure maximum tractive effort under all conditions. The roller shall be the current commercial production model of a standard product, and except for product improvements which have been incorporated on the commercial production units, shall have been marketed for at least 1 year prior to the opening of technical proposals or bids. In addition to the above, the roller shall be complete with all components furnished as standard equipment that are customarily offered with a roller of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the the roller to function reliably and efficiently in a sustained operation. The roller shall comply with all applicable Environmental Protection Agency (EPA) and Department of Labor (OSHA) requirements in effect on date of manufacture.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

* 3.3.1 Characteristics. The roller shall conform to the following characteristics:

| | |
|---|---|
| Weight, Maximum (pounds (lb)) 1/ | 30,000 |
| Weight, Minimum (lb) 1/ | 20,000 |
| Height, Maximum (inch (in)) 2/ | 102 |
| Width, Maximum (in) | 106 |
| Length, Maximum (in) | 240 |
| Drum Length, Maximum (in) 3/ | 84 |
| Steering Articulation (degrees in each direction) | 27 |
| Travel Speed, Miles Per Hour (mi/h) | |
| Low Range, Minimum | 0-4 |
| High Range, Minimum | 0-8 |

3.3.2 Engine. The engine for the roller covered by this specification shall be a liquid cooled diesel engine with sufficient power and speed rating so that operation of the roller under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer. The engine shall be mounted on rubber isolators to reduce noise and vibration. The engine shall be capable of operating and meeting the performance specified herein on diesel fuel conforming to VV-F-800 and JP-5 turbine fuel conforming to MIL-T-5624.

* 3.3.2.1 Starting system. The engine shall be furnished with a 12-volt electric starting system with sufficient batteries to crank the engine under all conditions specified herein.

3.3.2.2 Engine lubrication system. The engine lubrication system shall provide lubrication for all roving engine parts for continuous operation. The lubrication system shall be a full pressure type system. The roller shall have oil drain provisions for draining the engine crankcase. A bayonet gage or dipstick which is calibrated for determining crankcase oil level shall be provided.

1/ Weight shall include fully serviced and fueled roller with padfoot drum module, pintle hitch, tires as specified and furnished as an integral part of the roller and roll-over protection structure, but shall riot include operator.

2/ Without roll-over protection structure; roll-over protection structure required but must be detachable.

3/ Maximum length applies to both smooth and padfoot type drum modules.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

3.3.2.3 Cooling system. The liquid cooling system consisting of radiator, air shroud, fan-water pump and thermostat shall be of adequate capacity to permit operation under continuous operating conditions at an ambient temperature of -25°F to +110°F. The above is exclusive of supplementary radiator shroud and fan cooling capacity necessary for hydraulic cooling should the engine cooling system also be used for that purpose.

3.3.2.4 Air cleaner. The engine, shall have a dry type filter, the capacity of which shall be in accordance with the engine manufacturer's recommendation. A visual warning device showing when service should be performed shall be provided.

3.3.2.5 Exhaust. The engine exhaust shall be under hood, or otherwise protected, to preclude water entering the engine during high wind rain storms. A rain cap or hat is not considered adquate under these conditions.

3.3.2.6 Starting aid. The engine shall be provided with a pressurized container 20-ounce capacity measured shot ether cold weather starting aid. The control for the starting aid shall be mounted on the instrument control panel and shall be operable with a single mittened hand.

* 3.3.2.7 Ignition switch. The engine shall be provided with an on-off switch conforming to MS39060 Type III to operate the gages and start the engine in lieu of the key type ignition normally provided.

3.3.3 Frame. The frame shall be constructed of structural steel sections or steel plates with members joined using one of the arc welding processes. Frame shall be reinforced to prevent distortion by utilizing cross braces, gussets, or a combination of both. Frame shall be hinged to facilitate steering. Hinge shall be located at or near the center of the roller and on the longitudinal axis of the frame. Frame shall be mounted on two drive wheels on rear and a single vibrating drum on front.

* 3.3.4 Propulsion system. Power from the engine shall be transmitted to the drive wheels by a hydrostatic drive or a selective gear transmission having not less than two speed ranges and a nonslip type differential, if sliding gear transmission is provided. The drum/frame modules provided shall also be equipped with hyrostatic drive. The roller shall be equipped with a single forward/reverse speed control which shall provide infinite speed selection without affecting the vibrations per minute of the vibration system. A separate lever shall be supplied for gear selection for the selective gear transmission? if provided. The propulsion system shall provide for dynamic braking. Equipped with the smooth drum, the roller shall operate at any speed from 0.5 mi/h to 8 mi/h while traveling on a level paved surface.



Source: https://assist.dla.mil -- Downloaded: 202640-20T22:43Z
Check the source to verify that this is the current version before use.

3.3.5 Steering system. The steering system shall provide for full power hydraulic steering of the articulated type, with horizontal oscillation of the drum provided at the steer pivot point.

* 3.3.6 Electrical system. The electrical power for the roller shall be supplied by a sufficient number of batteries for the low temperature operation specified herein. The system shall be designed for 12-volt negative ground.

3.3.6.1 Batteries. Batteries shall be 6 TN in accordance with MS35000-3. Batteries shall be furnished charged and dry; electrolyte is not to be provided.

* 3.3.6.2 Slave receptacle. The electrical circuit shall incorporate a slave receptacle conforming to MIL-C-62122 for starting the engine from auxiliary sources. The receptacle shall be located in a position accessible to a person standing on the ground.

* 3.3.6.3 Lighting system. The lighting system shall include two forward and two rearward directed sealed beam floodlights. Lamp housings shall be provided with adjustable mounting brackets and shall be positioned so the lights do not shine in the operator's eyes when in his normal operating position. All wiring shall be protected against wearing and weathering and shall be so secured that no section exceeding 3 feet is unsupported. Grommets shall be used where wiring passes through structural steel sections and sheet metal to prevent chafing of the cable.

3.3.7 Brake system. In addition to the dynamic braking, adequate hydraulic service brakes and an adequate hand or foot operated emergency/ parking brake shall be provided. Brake system performance shall be in accordance with SAE J1136.

3.3.8 Hydraulic system. Engine driven hydraulic pumps shall supply the power for all roller hydraulic functions. The hydraulic system shall be complete with all operating accessories, including pumps, reservoir, hydraulic cylinders, valves, pressure gages, pressure relief valves, filters, hoses, piping and fittings. Hydraulic circuits shall be designed to promote simultaneous operation of all functions without interference. Main relief valve settings shall be provided to protect hydraulic components.

3.3.8.1 Hydraulic fluid. The hydraulic fluid used in the system shall conform to MIL-L-2104. All components and materials shall be compatible with these oils throughout an oil operating temperature range of -25°F to +225°F.



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

3.3.8.2 Hydraulic relief valves. Each hydraulic circuit shall be supplied with main hydraulic relief valves set for the working pressure of that circuit. Secondary relief valves shall also be supplied where necessary to protect circuit.

* 3.3.8.3 Hydraulic oil reservoir. The hydraulic reservoir assembly shall include a magnetic drain plug, atmospheric breather filter, fill cap, level indicator and 100-mesh outlet screen or equivalent. Reservoir shall be of sufficient size and design to absorb volume changes within the system and provide for fluid deaeration and makeup.

3.3.8.4 Hydraulic oil cooler. A hydraulic oil cooler shall be furnished to dissipate any heat buildup in the hydraulic system when operating the roller under continuous operating conditions in ambient temperatures up to 110°F. Air or liquied type cooler shall be installed in a low pressure return line. Protection from surge pressures shall be provided.

* 3.3.8.5 Hydraulic oil lines. High pressure oil lines shall be steel tubing whenever practical and shall conform to SAE J525. Hydraulic hose assemblies shall also conform to applicable SAE standards and shall meet the age requirements specified in 3.2.

* 3.3.8.6 Hydraulic hoses disconnect. Those hoses required for operating the two interchangeable vibratory drums/drum and frame assemblies shall be equipped with disconnectable fittings. Dust caps and plugs shall be attached to the fitting at the point of disconnect. The caps and plugs shall be provided for both roller and drum/drum and frame assembly hoses.

3.3.9 Vibration system. A variable frequency vibrating mechanism shall be provided. The vibration system shall be of the direct hydraulic drive or a combination hydraulic/belt drive driven by the roller engine. The belt drive shall be restricted to that area between the hydraulic motor and the vibrator shaft drive head. The vibrating mechanism shall have a minimum frequency of not less than 1,100 vibrations per minute with a range of adjustment of not less than 350 vibrations per minute. Operating at the manufacturer’s recommended maximum continuous frequency, the centrifugal force generated by the vibrating mechanism shall be not less than 23,000 pounds. The vibratory force generated by the vibrating mechanism shall be effectively isolated from the rest of the roller. The vibrating mechanism shall be contained within the drum. All rotating shafts shall be mounted on heavy-duty bronze or antifriction type bearings. Controls, located at the operator’s station, shall control the engagement, disengagement and frequency of the vibrating mechanism.



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

3.3.10 Smooth drum. The roller shall be equipped with a smooth type drum. The dimensions of the drum shall be as specified in 3.3.1. The shell and heads of the drum shall be constructed of steel plates. The drum shall be furnished with an axle assembly. The axle shall be mounted on antifriction type or heavy-duty type bearings.

* 3.3.11 Padfoot drum. A padfoot drum shall also be furnished with the roller. The padfoot drum shall be interchangeable with the smooth drum. The interchangeability feature must be capable of being accomplished on the job site with either normal tools or special equipment, which if required, shall be provided with each roller, in two hours with a crew of three. The padfoot drum shall be furnished with an independent axle assembly mounted on antifriction type or heavy-duty bronze type bearings. The quantity and type of padfoot provided on the drum shall be the manufacturer's standard.

* 3.3.12 Drum changeover equipment. The roller shall be furnished with any special equipment required for changing drums in the field without use of heavy equipment. Any special equipment provided shall be capable of being stored in the tool box specified in 3.3.18.

3.3.13 Drum cleaner. A replaceable and adjustable cleaner shall be provided for each type of drum provided with the roller. The cleaners shall extend over the full length of their respective drum and shall prevent accumulation of material on the drum when the roller is propelled in either the forward or reservse direction.

* 3.3.14 Fuel tank. The fuel tank shall have a capacity sufficient to provide 8 hours operation. Such operation shall be recognized as a combination of roving, compacting or other normal functions resulting in maximum fuel consumption. The fuel tank filler cap shall be an approved diesel service cap capable of accepting a 1-3/4 inch refueler nozzle. The tank shall conform with OE5A0.AF(7813) Factory Mutual service requirements.

* 3.3.15 Tires and rims. Tire and rims sizes shall conform to the Tire and Rim Association's recommendations for the type and size of tires furnished. Standard traction-type tires as normally provided on the commercial production units shall be provided. Tires shall be of a rated capacity at least equal to the load imposed measured at each wheel at the ground.

* 3.3.16 Operator's station. The roller shall be provided with an operator's station furnished with controls, an instrument panel and an operator's seat. All controls necessary to operate the roller shall be located at the station. The instrument and control panel located at the operator's station shall be fitted with, but not necessarily limited to the following:



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

(a) Liquid quantity fuel indicator
(b) Lubricating oil pressure indicator
(c) Cooling liquid temperature indicator
(d) Tachometer
(e) Hour meter
(f) Battery charging indicator
(g) Vibrations per minute indicator
(h) Manual speed control
(i) Normal shutdown device
(j) Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.

All instruments and controls shall be easily accessible and clearly visible, during day or night opeations, to the operator in his normal sitting position. All controls and instruments shall be clearly identified for their intended operation and function with permanently affixed identification plates made of corrosion-resistant material. The operator's seat shall be adjustable and shall be covered with a durable, waterproof covering. Steps and ladder rungs and handholds shall be furnished for easy access to the station. The station's deck and steps shall be provided with an antiskid surface. Station shall be isolated from vibrations generated by the vibrating mechanism.

* 3.3.17 Special tools. Any special tools required for equipment operation and operator maintenance shall be provided with each roller.

*3.3.18 Toolbox. The roller shall be provided with a closed compartment or toolbox to carry tools and accessories required for operator maintenance. The compartment or toolbox shall have a hinged lid with a padlock closing device less lock and key. The toolbox shall be securely fastened to the roller in a readily accessible and protected position.

* 3.3.19 Lifting and tiedown attachments. The roller shall be equipped with lifting and tiedown attachments conforming to type III of MIL-STD-209. Attachments shall be indicated by a transportation plate conforming to MIL-P-514.

* 3.3.20 Towing attachments. Towing attachments, if offered as standard or optional equipment, shall be provided.

* 3.3.21 Pintle hook. A rear-mounted pintle hook conforming to MS51118 shall be provided. The pintle hook shall be located on the vertical center line of the roller not less than 31 inches nor more than 37 inches above ground level. The pintle hook mounting shall be of sufficient strength to withstand, without permanent defamation or damage, the drawbar pull induced when the pintle hook is restraining forward motion at maximum drawbar pull.

* 3.3.22 Lubricants. All moving parts subject to wear or areas requiring



Source: https://assist.dla.mil -- Downloaded: 2026l0-20T22:43Z
Check the source to verify that this is the current version before use.

lubrication shall be provided with a means of lubrication. The roller shall be designed to operate efficiently and satisfactorily when lubricated with standard military lubricants conforming to MIL-L-2104 and MIL-46167 for engine oils; MIL-G-10924 for automotive grease; MIL-L-2105 for gear oil, and MIL-L-2104 for the hydraulic system oil and the transmission/torque converter oil. Each roller shall be serviced with military lubricants.

* 3.3.23 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures, that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded on insulated to the extent necessary to eliminate the hazard. All walking surfaces shall be provided with an antiskid surface. Operator visibility shall be adquate to permit safe operation in forward and reverse travel. All platforms, steps and handholds shall permit unhampered and nonhazardous entrance and exit from the roller. If the noise level in the area occupied by the operator exceeds 85 decibels (dB) (A), as measured in accordance with SAE J919b, a warning plate containing the following shall be permanently affixed to the roller in a conspicuous protected location:

"WARNING: EAR PROTECTION REQUIRED WHEN EQUIPMENT IS OPERATING."

The plate shall have a yellow background with black lettering and shall be of corrosion resistant material. The roller shall comply with the latest SAE requirements applicable to vibratory rollers to include roll-over protection structure. Roll-over protection structure must be detachable,

3.4 Reliable characteristic requirement. Each roller produced shall perform as reliably as the first article.

3.5 Construction.

3.5.1 Steel fabrication. Steel used in the fabrication of the roller shall be free from kinks and sharp bends. The straightening of material shall be done by methods which will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Reentrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.5.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers, or other positive locking shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient



Source: https://assist.dla.mil -- Downloaded: 2026l0-20T22:43Z
Check the source to verify that this is the current version before use.

strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.5.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.5.4 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack or intermittent welds for strength will not be permitted. Weld penetration shall provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall conform to the procedures as prescribed by either the AWS or the ASME for the type of welding operation to be performed.

3.5.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings, and otherwise to the standards of good commercial practice.

3.5.6 Castings. Castings shall be sound and free from defects which might render the casting unsound for use.

* 3.5.7 Threaded parts. Threaded parts shall be in accordance with the applicable SAE fastener standard or DIN grade 8.8. All hex bolts and capscrews shall be SAE grade 5 or 8 and shall be so marked.

3.6 Maintenance characteristic requirement. The roller design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field. Access doors shall be capable of being held in the open position. Maintenance shall be accomplished with the use of conventional general purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the roller.

3.7 Transportability. The roller shall be designed and constructed so that it can be transported by air, rail, highway and water transport

3.8 Performance. The roller shall withstand without damage operation on unimproved terrain, towing by another vehicle, and storage as specified herein.

* 3.8.1 Smooth drum compaction. The roller shall be capable of compacting at least 1100 compacted cubic yards per hour of soil conforming to SAE J106 type GP Unified Soil Classification System at least 95 percent



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

T99 AASHTO in Compacted lifts of not less than 12 inches when employing the smooth drum/frame module and prime rover combination.

3.8.2 Padfoot drum compaction. The roller shall be capable of compacting at least 825 compacted cubic yards per hour of soil conforming to SAE J106 type CL Unified Soil Classification System to at least 95 percent T99 AASHTO in compacted lifts of not less than 10 inches when employing the padfoot drum/frame module and prime mover combination.

* 3.8.3 Travel gradeability. The roller shall, at a minimum, be capable of negotiating a 20 percent grade over material offering a rolling resistance of 150 pounds per ton or greater for a total effective grade of 27.5 percent.

3.8.4 Temperature. Without using auxiliary power from external sources, the roller shall be equipped to start and operate under continuous operation conditions in not more than 30 minutes after engine start in any ambient temperature from -25°F to +110°F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures (sump and gallery) and component manufacturer's hydraulic fluid temperature limits. The roller shall not be damaged by daily storage at the above temperatures or for 30-day period at temperature as low as -65°F or as high as +160°F.

3.8.5 Altitude. The roller shall be powered with an engine which will provide the power required to perform the requirements specified herein at 5,000 foot altitude.

3.8.6 Rain and wind. The roller shall be designed to withstand extended periods of high winds and heavy rains during operation and storage. Under these conditions, there shall be no evidence of leakage into the intake filter, exhaust system, engine crankcase, hydraulic or other reservoirs and the engine shall start in 2 minutes after such exposure.

3.9 Electromagnetic interference. The electromagnetic radiation limits of the roller shall conform to MIL-STD-461; Class IIIC.

3.10 Dimensions. Dimensions shall be as specified in 3.3.1.

3.11 Weight. Weight shall be as specified in 3.3.1.

3.12 Cleaning, treatment and painting. The surfaces normally painted in good commercial practice shall be cleaned, treated and painted as specified in MIL-T-704, type A. The finished coat shall be in accordance with MIL-E-52798 (forest green).

* 3.13 Identification markings. Each individual roller and each separate attachment removable for either surface or air transportability or other purpose shall be identified in accordance with MIL-STD-130. A.U.S. Marine



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

Corps nameplate shall be permanently attached in a conspicuous location. The nameplates will be furnished upon request, by the Commandant of the Marine Corps (Code LMB), Headquarters U.S. Marine Corps, Washington, D.C. 20380. Data applicable to blank spaces 1, 3, 11, and 12 will be furnished by the procuring activity. The contractor shall be required to imprint the data in all blank spaces. All other plates, other than the nameplate, shall be provided by the manufacturer and shall conform to MIL-P-514, type I, grade A, class 1. Each plate shall be attached by screws, drive screws, bolts or rivets in a conspicuous, protected location.

3.13.1 Instruction plates. Each roller shall be equipped with instruction plates, including warnings and cautions, suitably located describing any special or important procedures to be followed in-operating, servicing, or disassembling the roller. Plates shall be of a material which will last and remain legible for the life of the roller, and shall be securely affixed thereto with brass screws of not less than 1/8-inch diameter or by other means.

3.13.2 Registration data. The registration number assigned to each roller by the contracting officer shall be affixed on the roller by the contractor using adhesive backed decals, nonreflective, conforming to type II, class I of MIL-M-43719. Registration numbers shall be applied on right and left sides, as well as the front and rear of the machine in minimum 2-inch characters. Color shall be black.

3.13.3 Stenciling. The gross weight of each roller shall be stenciled on each side of the roller in such a manner as to be readily discernible to dock and military personnel. The prescribed tire pressure shall be stenciled on the carrier fenders by block or stencil-type letters not more than 1-inch high. Decals conforming to type II, class I of MIL-M-43719 shall be utilized. Color shall be black.

3.14 Government-furnished property. The Government shall furnish a, rifle mounting kit which shall be mounted in a location accessible to the operator. The location shall be approval by the contracting officer.

3.15 Workmanship. All parts, components and assemblies of the roller, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves tile right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)
(b) Production (see 4.4)
(c) Preparation for delivery (see 4.5)

4.3 First article. First article inspection shall be performed on the first production item. This inspection shall include the examination in 4.6 and the tests in 4.7. The first article may be a standard production item from the supplier's current inventory, provided the roller meets the requirements of this specification and is representative of the design, construction and manufacturing technique applicable to the remaining rollers to be furnished under the contract.

4.4 Production. Each roller produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warm-up period as recommended by the manufacturer. Each roller shall be subjected to examination in 4.6. The roller shall be given a run-in test and all controls shall be operated a sufficient number of times to assure that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction.

4.5 Packaging. The preservation, packaging, packing, and marking of the roller shall be inspected to verify conformance to the requirements of Section 5.

4.6 Examination. Each roller shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with requirements specified herein or any necessary redesign or modification following failure to meet requirements specified herein shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any requirements specified herein or presence of one or more defects preventing or lessening efficiency shall constitute cause for rejection.

4.7 Tests. Failure of the roller to pass any of the following tests or failure of any component during testing shall be the basis for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

4.7.1 Control function. All operating controls and safety devices specified shall be identified and operated. Inability to meet the requirements specified herein shall constitute failure of this test.

4.7.2 Characteristics. The characteristics specified in 3.3.1 shall be verified.

4.7.3 Engine horsepower. The manufacturer shall certify that the . engine manufacturer has approved the engine for application as specified herein and shall furnish (1) a certified copy of the dynamometer test performed on a sample engine per SAE J816b, (2) a performance curve of fully equipped engine as installed in the roller, and (3) derating factors applicable when the engine is operated at altitudes above those specified in 3.8.5 and ambient temperatures above 85°F.

* 4.7.4 Speed and frequency. Roller shall be tested with the smooth drum on a course having a firm, smooth, level surface. The test course shall have a straight live length of not less than 200 feet. At each end of the course, sufficient area shall be provided to allow the roller to turn around with stopping its forward motion. One end of the course shall be designated for right hand turns only, the opposite end for left had turns only. During the test, the operator shall adhere to the turn designations. To establish proof of conformance to the requirements of 3.3.4 and 3.3.9, the roller shall be tested for not less than 20 minutes in the low speed range specified in 3.3.4, utilizing the full range of the frequency range specified in 3.3.9.

* 4.7.5 Maneuverability. Test course shall have a relatively level grade with a rouqh surface. Degree of roughness shall be sufficient to demonstrate the smoth drum's ability to oscillate to its maximum limit. Course length shall be not less than 200 feet. Width shall be sufficient to allow roller to turn around without reversing the traction drive. The operator shall maintain the highest possible travel speed as dictated by the roughness of the course and without jeopardizing good safety practices. The roller shall perform the following maneuvers for not less than 20 minutes each with the smooth drum.

(a) Figure-eight turns using the shortest possible turning radius and with the vibrating mechanism operating at a frequency of not less than 1,400 vibrations per minute.
(b) Travel back and forth in forward direction with vibrating mechanism operating at not less than 1,300 vibrations per minute.
(c) Travel back and forth in reverse direction with vibrating mechanism operating at not less than 1,200 vibrations per minute.

'Throughout the test, the roller shall be observed to verify compliance with the requirements specified in 3.3, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.3.8, 3.3.9, 3.3.13, and 3.8.4.

4.7.6 Brake system tests. The service and parking brakes shall be tested in accordance with SAE J1136. Inability to perform as specified shall constitute failure of this test.



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

4.7.7 Hydraulic hose test. Hydraulic hoses and fittings shall be bench tested to determine conformance to 3.3.8.5, except that certification by the manufacturer, or certified copies of test reports on identical items previously tested for identical pressures may be accepted.

* 4.7.8 Drum changeover test. The drum changeover equipment specified in 3.3.12, if required, shall be demonstrated to establish proof of conformance to the requirement of 3.3.11 and 3.3.12.

4.7.9 Padfoot drum test. The padfoot drum shall be interchanged for the smooth drum and the roller equipped with the padfoot drum shall be subjected to the manueverability test as specified in 4.7.5.

4.7.10 Lifting and tiedown attachments test. The roller shall be tested to verify that the attachments conform to the requirements specified in 3.3.19. Failure of any component, or breakage, distortion or deformation shall constitute failure of this test.

4.7.11 Pintle hook test. The pintle hook shall be tested by exerting the maximum drawbar pull on it. Any distortion or breakage shall constitute failure of this test.

* 4.7.12 Sound level test. The roller shall be tested to determine conformance with 3.3.23 and SAE J919b. Roller shall meet SAE J919B or have warning specified in 3.3.23.

* 4.7.13 Compaction test. The contractor shall furnish actual data ("soil-bin" analysis not acceptable) to prove that the roller is capable of compacting at least 1100 compacted cubic yards per hour of soil conforming to SAE J106 type GP Unified Soil Classification System to at least 95 percent T99 AASHTO in compacted lifts of not less than 12 inches when employing the smooth durm. The contractor shall also furnish actual data ("soil-bin" analysis not acceptable) to prove that the roller is capable of compacting at least 825 compacted cubic yards, per hour of soil conforming to SAE J106 type CL Unified Soil Classification System to at least 95 percent T99 AASHTO in compacted lifts of not less than 10 inches when employing the padfoot drum. Soil reports shall be furnished which include the gradation of the soils and the densities achieved. The soil reports shall also include the type of density test equipment used, i.e., sand cone, nuclear density meter, etc. A narrative description of the compaction operation shall be furnished. The narrative description shall include the actual cubic yards of fill compacted, amounts of compaction time, compacted lift thickness, how the soil was placed and a list of water distribution equipment used, if any.

4.7.14 Productivity test. Operate each roller/drum module combination over a natural soil subgrade which has been compacted to a density of at least 90 percent T99 AASHTO. For evaluation of the smooth drum module, the fill material shall correspond to SAE J106 type GP of the Unified Soil Classification System. For evaluation of the padfoot drum module, the fill material shall correspond to SAE J106 type CL of the Unified Soil Classification System. The compacted fill width shall be 31 feet wide with the actual width of the fill being a



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

minimum of 33 feet. The fill may be placed prior to the start of compaction operation, during compaction operations of the contractor may elect to till the natural soil in lieu of placing fill material. Water distribution equipment may be used as required. The type GP material to be compacted shall be of sufficient thickness to give a compacted lift of at least 12 inches. The type CL material to be compacted shall be of sufficient thickness to give a compacted lift of at least 10 inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D-1557-70, method optional. The Government shall witness any laboratory testing. When the contractor feels that compaction of 1100 compacted cubic yards of GP material, and of 825 compacted cubic yards of type CL material for the second test, has been accomplished to the required density of 95 percent T99 AASHTO, compaction operations may be stopped. Time used in density testing shall not be counted as compaction time. Samples shall be taken at points along the test strip using the pattern shown on Figure 1 at intervals of 400 feet. In addition, another three samples shall be taken at random locations specified by the Government. Density testing shall be accomplished by any nuclear density meter. The nuclear density tester shall have been calibrated against the soil to be compacted. The Government shall witness such calibration. An achieved density of less than 95 percent T99 AASHTO shall constitute failure of this test. Sufficient measurements shall be taken to determine the in-place quantity of material compacted. The above procedure shall be repeated for a total of three times. An average compaction rate of less than 1100 compacted cubic yards per hour of type GP material with a minimum compacted lift thickness of 12 inches, and an average compaction rate of else than 825 compacted cubic yards per hour of type CL material with a minimum compacted lift thickness of 10 inches shall constitute failure of these tests.

* 4.7.15 Gradeability. Operate each roller/drum module combination on a minimum grade of 20 percent fill material having a rolling resistance of not less than 1500 pounds per ton (total effective grade 27.5 percent). The offeror may, at his option, operate each prime mover/drum module combination up a grade of at least 27.5 percent with a smooth surface. Inability to negotiate the grade with either work module under either condition specified above shall constitute failure of this test.

* 4.7.16 Wind and rain test. The wind and rain test shall be conducted as follows:

(a) TEST APPARATUS:

(1) Timing device.

(2) Measuring device (U.S. Weather Bureau type preferred).

(3) Spray fixture.



Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.

(b) TEST PROCEDURE:

(1) Subject the roller to a simulated rainfall of not less than 4 inches per hour. The similated rainfall shall impinge on the roller at an angle of 45 degrees, ± 5 degress, with a force equivalent to a 30 mi/h wind. The simulated rainfall shall disperse uniformly over the surfaces of the roller.

(2) Expose the top and left side of the roller to the rain described above for a period of 5 minutes with engine off. With rain still flowing start engine and allow engine to idle for 2 minutes.

(3) Repeat (b) with the top and front exposed.

(4) Repeat (b) with top and right exposed.

(5) Repeat (b) with top and rear exposed.

(6) Stop the rainfall and allow the roller to stand for not less than 30 minutes with the engine off. Drain a small quanitity of fluid from each fluid reservoir (except cooling system) and examine for presence of water.

(7) Check electrical components, gages, cab (if furnished), and the air cleaner for water entry and damage.

* 4.7.17 Test for electromagnetic interference characteristics. The roller shall be tested in accordance with MIL-STD-462 for Class III equipment. Inability to pass the test shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The complete roller shall be preserved and packaged in accordance with level A of MIL-R-3075.

5.1.2 Level C. The complete roller shall be preserved and packaged in accordance with the contractor’s standard practive.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The complete roller shall be packed in accordance with level A MIL-R-3075.

5.2.2 Level B. The complete roller shall be packed in accordance with level B of MIL-R-3075.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The complete roller shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules.

5.3 Marking. The rollers, attachments, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The roller described herein shall be used in military operations involving construction of roads, airfield complexes, open storage areas, hard stands for vehicle and equipment parks, repair of bomb damage on airfields, compaction of building sites and compacting asphalt. The roller shall be subject to the riggers of amphibious loading and unloading from naval vessels, as well as sea transport and other conditions inherent to military operations.

6.2 Ordering data. Procurement data should specify:

6.2.1 Procurement requirements:

(a) Title, number and date of this specification.

(b) Level of preservation, packaging and packing (5.1 and 5.2).

6.6.6 Contract data requirements. Data conforming to Data Item Description DI-E-4527, DI-M-4704 and DI-R-4802 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 First article. Any changes or deviations of production rollers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish rollers conforming to this specification.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsover for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - MC
Project No. 3895-N198



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:43Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-R-29385A(MC)

ROLLER, MOTORIZED: COMPACTOR, VIBRATORY ARTICULATED,DIESEL ENGINE DRIVEN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-LF-014-1802

Source: https://assist.dla.mil -- Downloaded: 202610-20T22:43Z
Check the source to verify that this is the current version before use.